

Air Compressor Monitoring

Using IRIV PiControl

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Chapter 1: Project Overview

In modern industrial environments, Programmable Logic Controllers (PLCs) are widely used for machine control, process automation, and data acquisition. Although PLCs are reliable and robust, their implementation often involves high hardware costs, proprietary software, and limited flexibility for IoT integration and remote monitoring applications.

With the advancement of Industrial Internet of Things (IoT) technologies, industries are increasingly seeking cost-effective and flexible alternatives that can perform data acquisition, communication, and monitoring functions while maintaining industrial-grade reliability. The IRIV PiControl is an industrial controller designed to support various industrial communication protocols, making it a potential alternative solution for selected PLC-based monitoring applications.

To demonstrate the capabilities of the IRIV PiControl, an Air Compressor Monitoring System was developed as a proof-of-concept project. Air compressors are critical assets in industrial facilities, and their operating conditions must be continuously monitored to support Condition-Based Maintenance (CBM) practices. CBM relies on real-time equipment data to detect abnormalities and predict maintenance requirements before failures occur.

The developed system integrates multiple industrial sensors including pressure sensor, temperature & humidity sensor, vibration sensor, pressure sensor, flow sensor, thermocouple and power meter through the Modbus RS485 communication protocol. The IRIV PiControl serves as the central controller that collects data from multiple devices. It will process the information and visualize the results through a Node-RED dashboard.

This project demonstrates that the IRIV PiControl can effectively communicate with multiple Modbus-enabled sensors simultaneously, perform industrial data acquisition tasks, and provide real-time monitoring capabilities. The successful implementation of this system highlights the potential of the IRIV PiControl as a flexible and cost-effective alternative for industrial monitoring applications traditionally performed by PLC systems.

Chapter 2: Objective

The objective of this project is to develop an air compressor monitoring system using the IRIV PiControl to demonstrate its capability as an industrial monitoring controller and potential alternative to conventional PLC systems. The system integrates multiple sensors through Modbus RS485 communication to collect, display and analyze real-time compressor operating data while providing alert notifications when abnormal conditions are detected. Other than that, it is supporting condition-based maintenance (CBM) and improving equipment monitoring efficiency for maintenance.

Chapter 3: Scope of Project

This project consists of two main components which is hardware and software development. The hardware component focuses on the integration of the IRIV PiControl with multiple industrial sensors through Modbus RS485 communication for real-time data acquisition. The software component focuses on the development of a monitoring dashboard, data processing, data logging, and alert notification functions to enable effective monitoring of air compressor operating conditions.

3.1 Hardware

The hardware implementation of this project consists of the IRIV PiControl and several industrial sensors used to monitor the operating condition of an air compressor. All sensors communicate with the IRIV PiControl through the Modbus RS485 protocol, allowing multiple devices to be connected and monitored simultaneously. The selected sensors measure critical parameters required for condition-based maintenance (CBM), including pressure, temperature, vibration, and flow rate.

Pressure Sensor



Figure 3.1: Pressure Sensor

Figure 3.1 shows the pressure sensor used in this project. The sensor is installed on the compressor airline to measure the compressed air pressure and temperature. The collected pressure data is transmitted to the IRIV PiControl via Modbus RS485 for real-time monitoring and analysis.

Temperature & humidity Sensor



Figure 3.2: Temperature Sensor

Figure 3.2 shows the temperature sensor used to monitor the operating temperature of the compressor. Temperature monitoring is important for detecting overheating conditions that may affect compressor performance and reliability. The sensor communicates with the IRIV PiControl through the Modbus RS485 network.

Vibration Sensor



Figure 3.3: Vibration Sensor

Figure 3.3 shows the vibration sensor installed on the compressor motor. The sensor measures vibration levels that can indicate mechanical issues such as bearing wear, shaft misalignment, or imbalance. Continuous vibration monitoring supports early fault detection and condition-based maintenance activities.

Flow Meter



Figure 3.4: Flow Sensor

Figure 3.4 shows the flow sensor used to measure the compressed air flow rate. Monitoring airflow helps evaluate compressor performance and identify potential leakage or abnormal operating conditions. The measured data is collected by the IRIV PiControl through Modbus RS485 communication.

Thermocouple Sensor



Figure 3.5: Thermocouple Sensor

Figure 3.5 shows the thermocouple sensor used in this project. The sensor is installed on the compressor motor to measure its operating temperature directly. Continuous temperature monitoring helps detect overheating conditions and provides valuable data for condition-based maintenance (CBM) by identifying abnormal temperature increases that may indicate potential equipment faults.

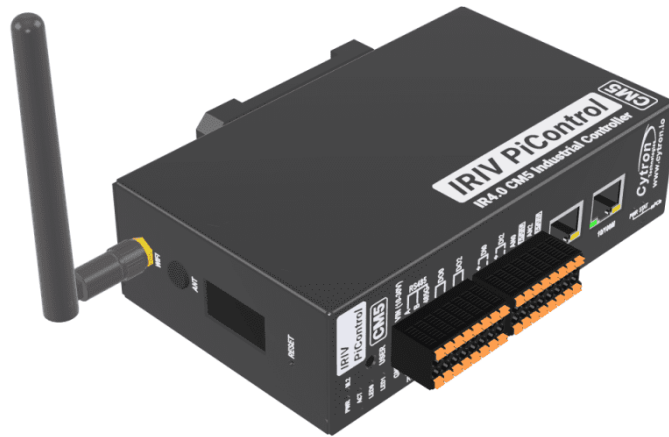


Figure 3.6: IRIV PiControl

Figure 3.6 shows the IRIV PiControl used as the main controller in this project. It acts as the central data acquisition unit by collecting data from all connected sensors, processing the information, and transmitting it to the monitoring dashboard. The device demonstrates the capability of handling multiple Modbus devices simultaneously and serves as a potential alternative to conventional PLC-based monitoring systems.

3.2 Software

The software implementation of this project focuses on data acquisition, visualization, and alert notification functions. Node-RED is used as the main development platform to communicate with the sensors through the IRIV PiControl, process the collected data, display real-time monitoring information, and generate alerts when abnormal operating conditions are detected. The software components developed in this project are described below.

Node-RED Flow

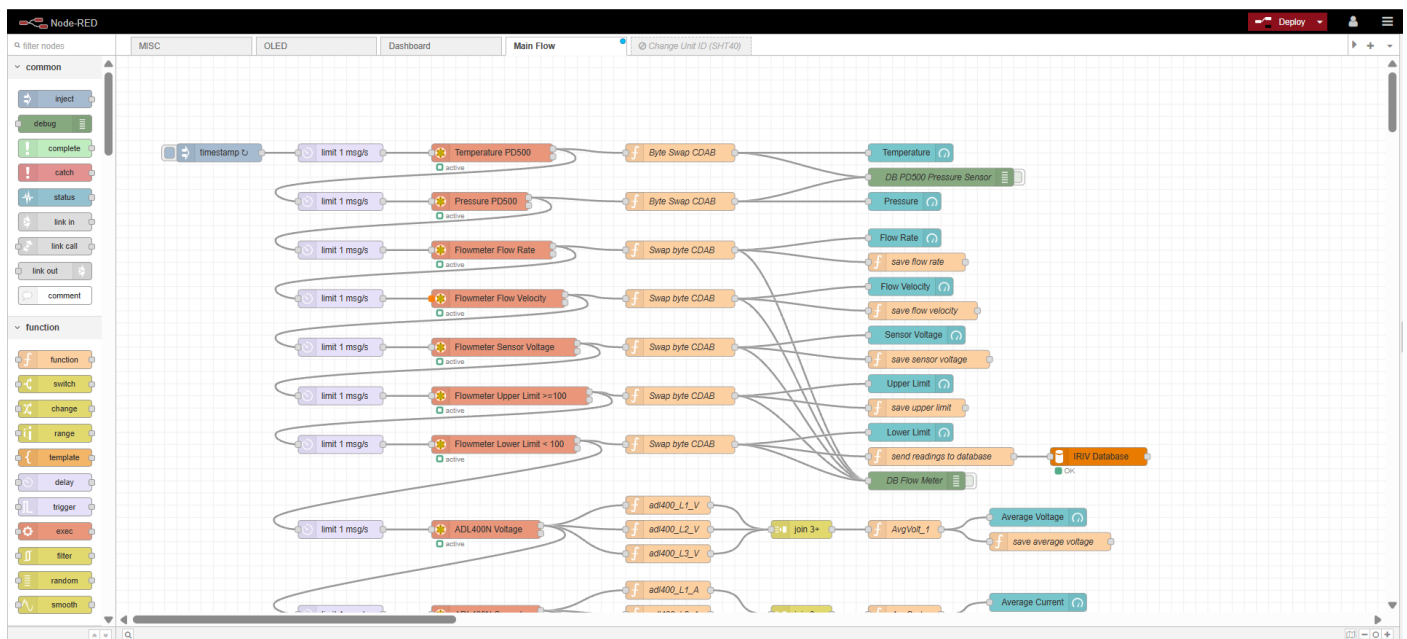


Figure 3.7: Node-RED Flow

Figure 3.7 shows the Node-RED flow developed for this project. The flow is responsible for communicating with multiple Modbus sensors, processing the received data, and distributing the information to the monitoring dashboard and alert system.

Monitoring Dashboard

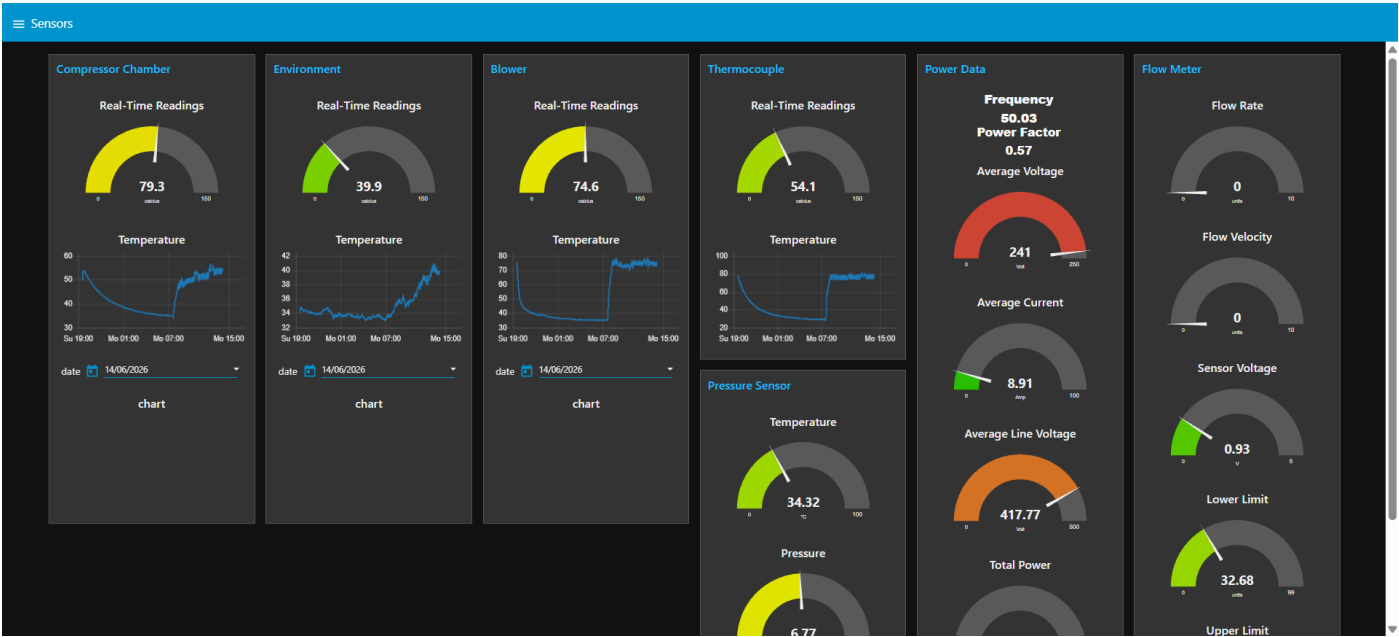


Figure 3.8: Monitoring Dashboard

Figure 3.8 shows the monitoring dashboard developed using Node-RED Dashboard. The dashboard provides real-time visualization of compressor parameters such as pressure, temperature, vibration, and flow rate, allowing users to monitor equipment conditions through a single interface.

Data Logging System

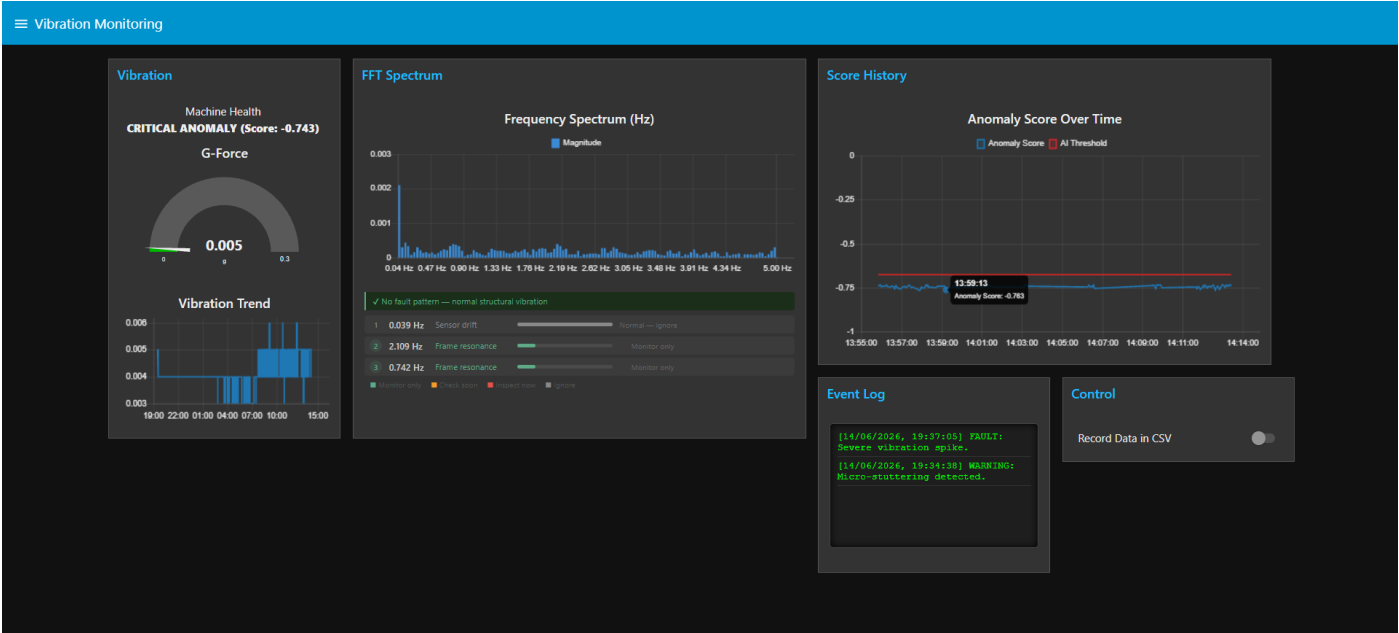


Figure 3.9: Data Logging System

Figure 3.9 shows the data logging function implemented in the system. Sensor data is recorded continuously for historical analysis and performance evaluation. The stored data can be used to identify trends and support maintenance planning.

Alert Notification System

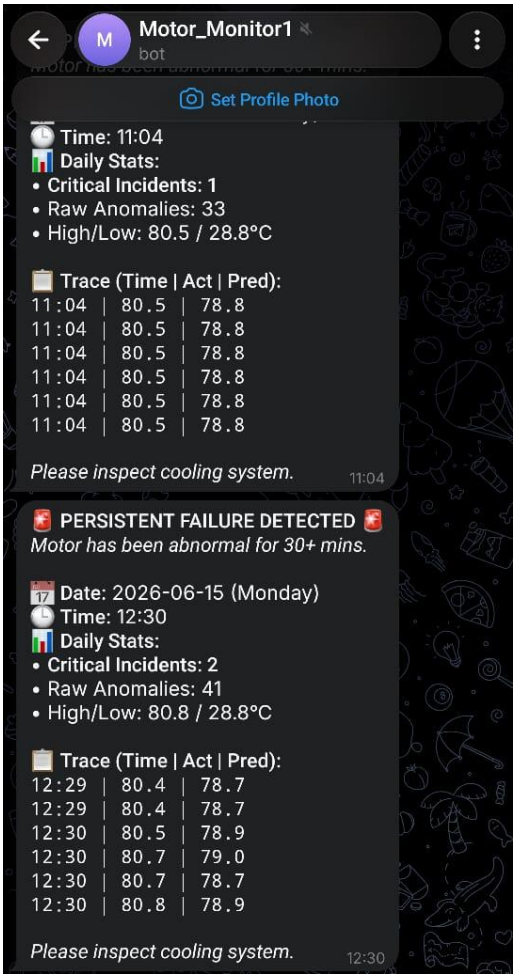


Figure 3.10: Alert Notification System

Figure 3.10 shows the alert notification system developed in this project. When sensor readings exceed predefined threshold values, the system automatically generates notifications to alert operators of potential abnormal conditions, enabling faster response and reducing the risk of equipment failure.

Chapter 4: Project Development

4.1 System Architecture

- Overall system block diagram

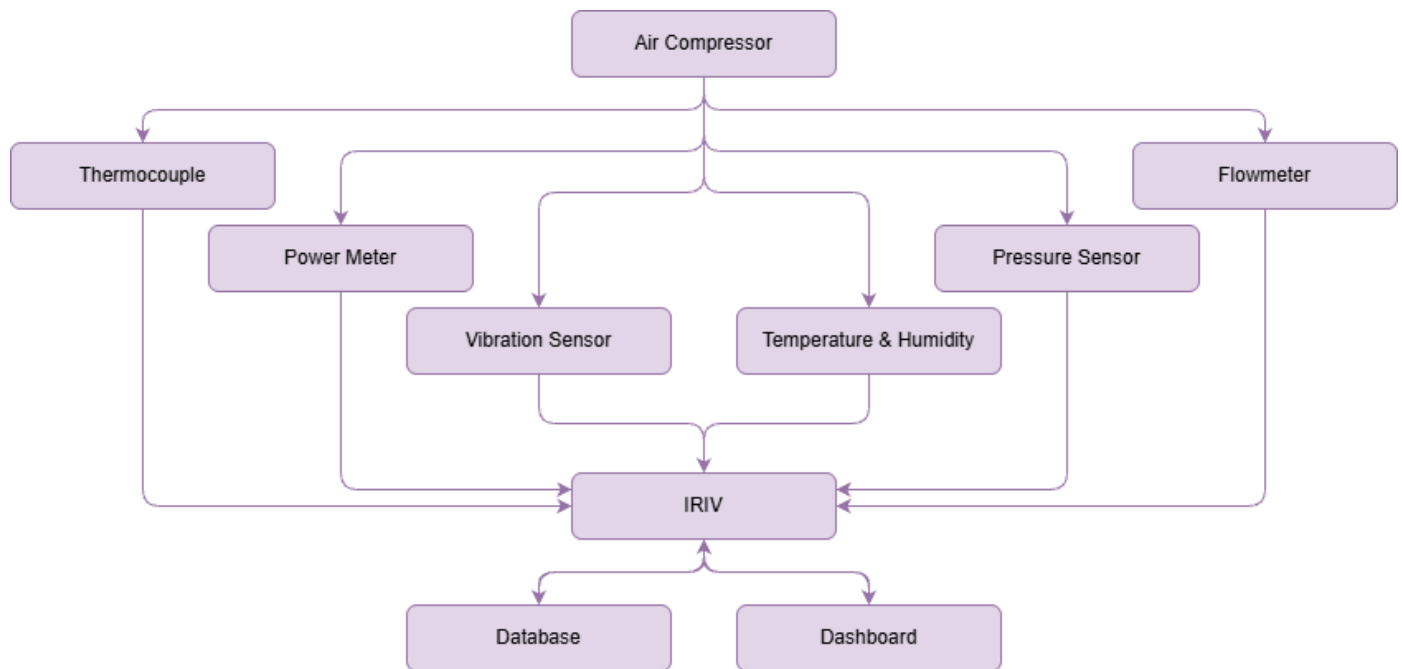


Figure 4.1.1: System Block Diagram

- Overall system flowchart

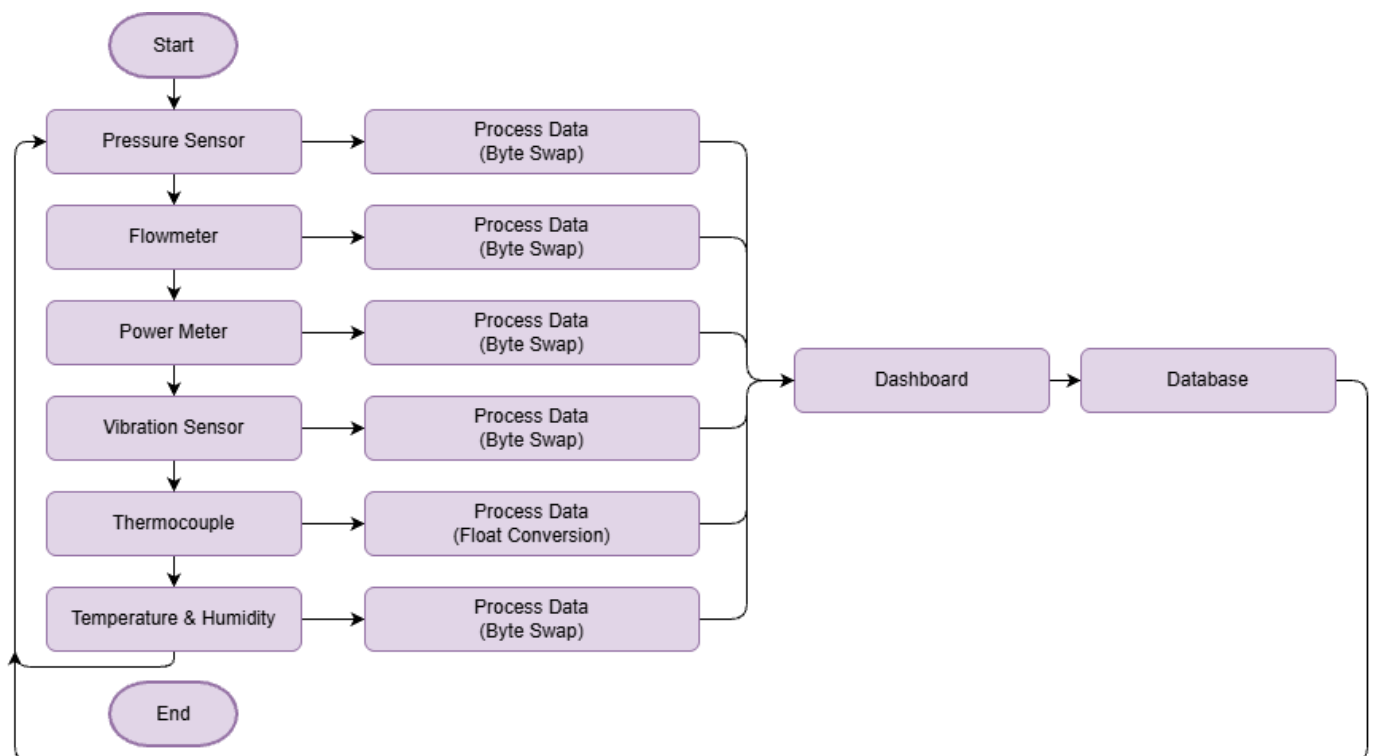


Figure 4.1.2: System Flowchart Diagram

4.2 Hardware Setup

- Sensor installation locations
- Wiring of the System

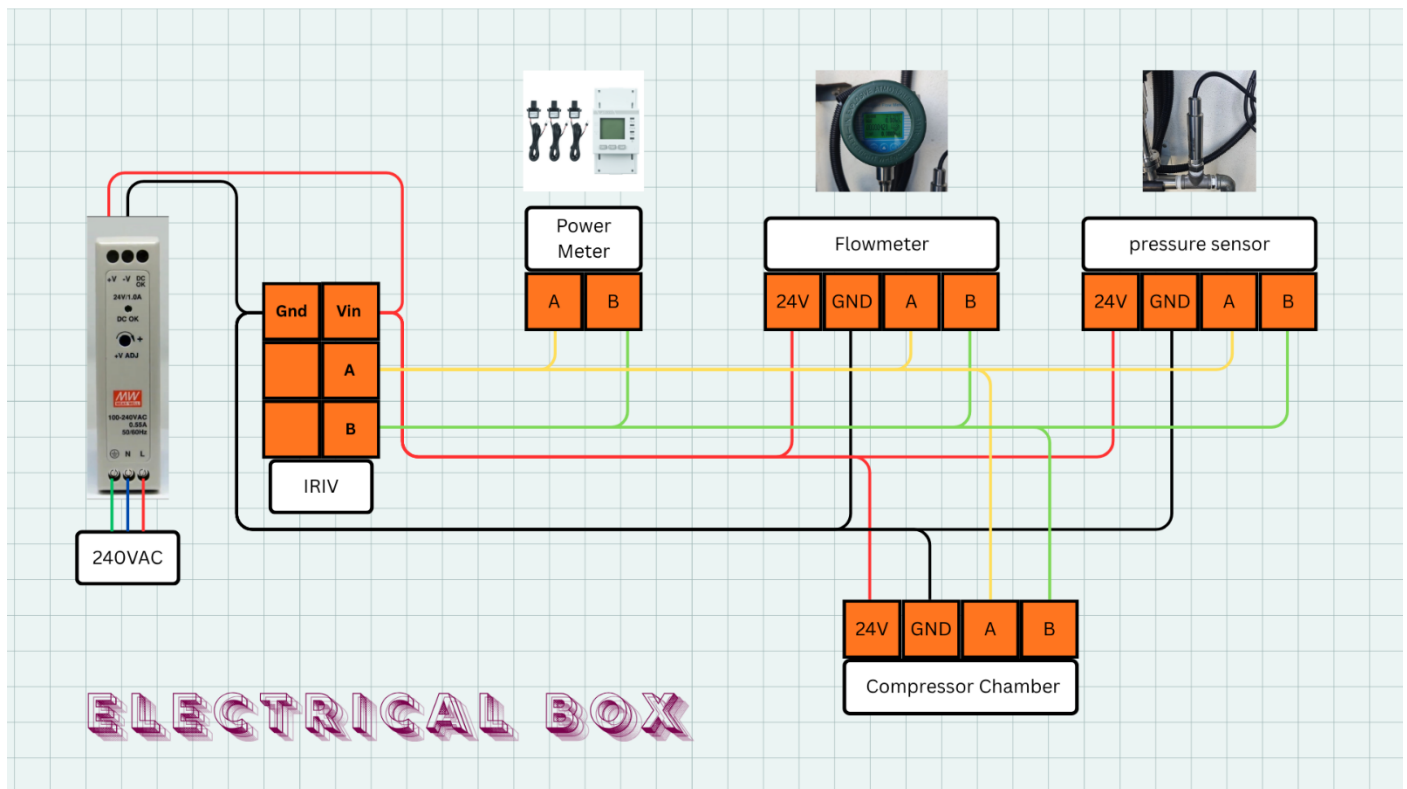


Figure 4.2.X: Hardware Wiring Diagram in Electrical Box

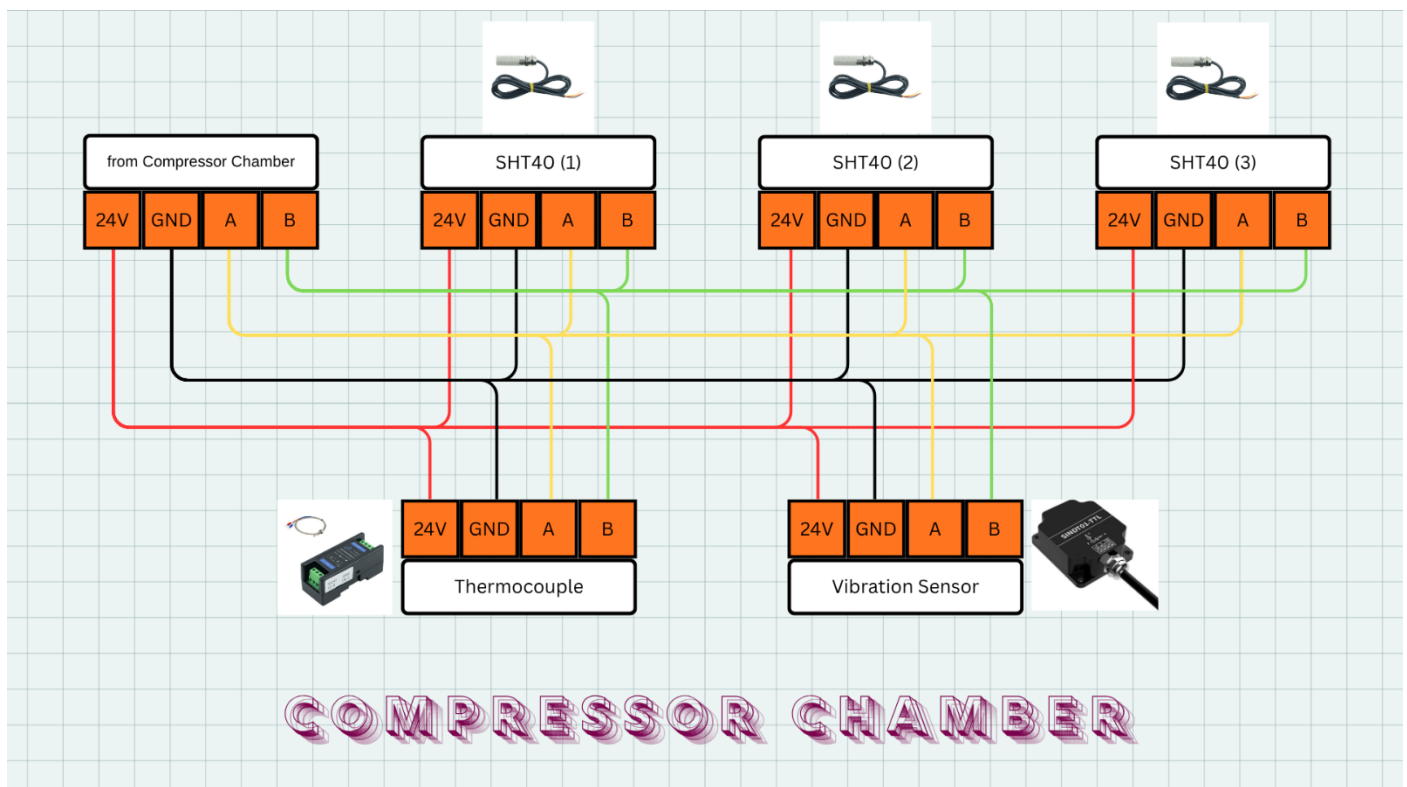


Figure 4.2.X: Hardware Wiring Diagram in Compressor Chamber

4.3 Modbus RS485 Integration

- Sensor slave ID configuration
- Baud rate settings
- Register mapping

4.4 Node-RED Development

- Modbus Getter configuration
- Data processing functions
- Data conversion 4 byte register address using CDAB format
- Flow development

Figure 4.4: Node-RED Flow

4.5 Dashboard Development

- Dashboard design process
- Gauges and charts
- User interface layout
- Real-time monitoring display

Figure 4.5: Compressor Monitoring Dashboard

4.6 Alert Notification Development

- Notification delivery using Telegram chat Method
- Threshold setting
- Alert logic
- Alert testing

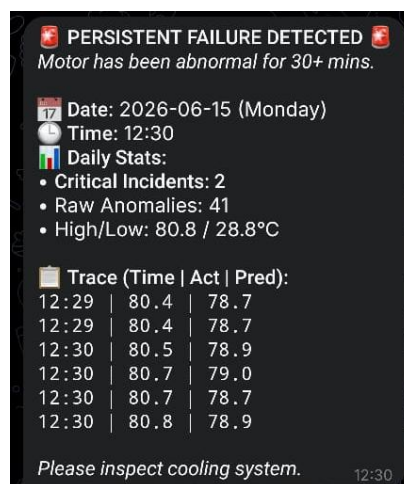


Figure 4.6: Alert Notification Example

4.7 System Testing and Validation

- Sensor reading verification
- Communication stability testing
- Dashboard functionality testing
- Alert system testing

Table 4.7: System Testing Results

Chapter 5: Data Analysis

5.1 Temperature Analysis

- Temperature readings collected from the thermocouple sensor
- Temperature trend over a specific period
- Normal operating temperature range
- Discussion of any abnormal temperature readings

Figure 5.1: Compressor Motor Temperature Trend

5.2 Pressure Analysis

- Pressure readings collected from the pressure sensor
- Pressure stability during compressor operation
- Analysis of pressure fluctuations

Figure 5.2: Compressor Pressure Trend

5.3 Vibration Analysis

- Vibration readings collected from the vibration sensor
- Comparison of vibration levels during different operating conditions
- Identification of unusual vibration patterns

Figure 5.3: Compressor Vibration Trend

5.4 Flow Rate Analysis

- Flow rate readings collected from the flow sensor
- Relationship between compressor operation and airflow
- Analysis of flow rate variations

Figure 5.4: Compressor Flow Rate Trend

5.5 Alert Notification Analysis

- Alert conditions configured in the system
- Example alerts generated during testing
- System response to abnormal conditions

Figure 5.5: Alert Notification Example

5.6 System Performance Analysis

- Modbus communication reliability
- Multi-sensor monitoring performance
- Dashboard responsiveness
- IRIV PiControl performance as a monitoring controller

Table 5.6: Summary of System Performance

Chapter 6: Barriers & Challenges

6.1 Challenges Before Project Implementation

- Understanding compressor monitoring requirements
- Selecting suitable sensors and communication protocols
- Learning the capabilities of the IRIV PiControl
- Obtaining and translating technical documentation and sensor datasheets

6.2 Challenges During Project Development

- Sensor installation and wiring
- Modbus RS485 communication configuration
- Assigning and managing Modbus Slave IDs
- Register address mapping and data interpretation
- Troubleshooting communication failures
- Alert notification implementation
- Sensor calibration and testing

6.3 Challenges After System Implementation

- Ensuring long-term communication stability
- Monitoring data accuracy and reliability
- Fine-tuning alert threshold values
- System maintenance and future scalability
- Integration with additional sensors or industrial equipment

6.4 Solutions and Improvements

- Methods used to overcome communication issues
- Improvements made to the dashboard and alert system
- Recommendations for future development
- Potential enhancements for industrial deployment
- Ni x habis lagi 6.4

Chapter 7: Conclusion

This project successfully developed an Air Compressor Monitoring System using the IRIV PiControl as the main monitoring controller. The system integrated multiple industrial sensors, including temperature, pressure, vibration, and flow sensors, through the Modbus RS485 communication protocol to acquire real-time operational data from the compressor.

A monitoring dashboard was developed using Node-RED to provide real-time visualization of sensor data, while an alert notification system was implemented to notify users when abnormal operating conditions were detected. The system supports Condition-Based Maintenance (CBM) by enabling continuous monitoring of critical compressor parameters and facilitating early fault detection.

The project demonstrated that the IRIV PiControl is capable of communicating with multiple Modbus devices simultaneously, performing data acquisition, processing sensor information, and supporting industrial monitoring applications without the need for a conventional PLC. Therefore, the project objectives were successfully achieved, and the results highlight the potential of the IRIV PiControl as a flexible and cost-effective solution for industrial monitoring systems.

Appendices

Appendix A: Sensor Datasheets

- Pressure Sensor Datasheet
- Thermocouple Sensor Datasheet
- Vibration Sensor Datasheet
- Flow Meter Sensor Datasheet
- Temperature & Humidity Sensor Datasheet
- Power Meter Datasheet

All of the Datasheet you may refer to this link:

https://github.com/ezrxleznisys/IRIV_COMPRESSOR/tree/54ea80c8eb00caa73b479614609cd08d3df99313/Resource%20%26%20References

Appendix B: Modbus Register Mapping

- Slave IDs
- Register Addresses
- Communication Parameters

For more detail table about Modbus RS485 Mapping and ID you may refer to the readme.md file through this link:

https://github.com/ezrxleznisys/IRIV_COMPRESSOR/blob/1c78b05dfbc7cb8255fe56db801a531d70b9d1c4/README.md

Appendix C: Node-RED Flow

- Debug flow is to check one by one sensor if failure

You may look for the debug flow on this link:

https://github.com/ezrxleznisys/IRIV_COMPRESSOR/tree/c1730405dfcd1357aab49dc63576060816460480/Debug%20Flow

- Complete Node-RED flow configuration

For full code in .json code format you may refer to this link:

Appendix D: Dashboard Screenshots

- Main monitoring dashboard
- Historical trend charts

Appendix E: Alert Notification Screenshots

- Alert messages

